

Fermi–Amaldi-Based Local Mixing Function

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I. MAIN TEXT

We propose a new local mixing function (LMF) for local hybrid functionals. This LMF is the ratio between the Fermi–Amaldi and the exact exchange energy densities. The (closed-shell) exact exchange energy density is given by [1]

$$e_x^{\text{exact}}(\mathbf{r}) = - \sum_{k,l=1}^{N/2} \int \frac{\varphi_k(\mathbf{r})\varphi_k^*(\mathbf{r}')\varphi_l^*(\mathbf{r})\varphi_l(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (1)$$

The Fermi–Amaldi [2, 3] energy density is expressed as

$$e_x^{\text{FA}}(\mathbf{r}) = - \frac{1}{2N} \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (2)$$

We can show that the proposed LMF satisfies the zero

to one boundary range:

$$e_x^{\text{exact}} \leq 0,$$

$$e_x^{\text{FA}} \leq 0,$$

$$e_x^{\text{exact}} \leq e_x^{\text{FA}},$$

thus

$$0 \leq \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} \leq 1. \quad (3)$$

For one-electron regions, $e_x^{\text{FA}} = e_x^{\text{exact}}$, therefore, the one-electron constraint is satisfied. A particular case of a one-electron limit, i.e., the asymptotic limit ($r \rightarrow \infty$) where the exact exchange must cancel out an excessive self-repulsion of the Coulomb potential, is also satisfied:

$$\lim_{r \rightarrow \infty} \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} = 1 \quad (4)$$

since both energy densities approach $-1/2r$ as $r \rightarrow \infty$.

Another important constraint, the high-density limit,

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- [1] V. N. Staroverov, Density-functional approximations for exchange and correlation, in *A Matter of Density* (John Wiley & Sons, Ltd, 2012) Chap. 6, pp. 125–156.
 - [2] E. Fermi and E. Amaldi, Le orbite ∞ s degli elementi, *Mem. Reale Accad. Italia* **6**, 117 (1934).
 - [3] P. Bénilan, J. A. Goldstein, and G. R. Rieder, The fermi-

amaldi correction in spin polarized thomas-fermi theory, in *Differential Equations and Mathematical Physics*, Mathematics in Science and Engineering, Vol. 186 (Elsevier, 1992) pp. 25–37.